Reading the folding history of a crease pattern from local histories at its vertices

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Abstract

A crease pattern records where a sheet was folded but not in what order. We give a method that recovers much of the order from the pattern alone, and test it on a corpus of published patterns. Its central object is the *local history* at a vertex: a sequence of folds, each through one line in the folded paper, that produces the creases at that vertex. We show that, apart from a collision test, local histories are characterised by two conditions on the creases made at one time: there is an even number of them, and any two are related by the product of reflections in the earlier creases lying between them. The product has a closed form, an alternating sum of directions, so the condition is linear once the set of earlier creases is fixed. Creases that were made and later erased are recovered as cancelled segments, which must themselves have local histories wherever they cross the pattern; moves that fold several lines at once are admitted at a cost. Requiring one order of all creases to be consistent with a local history at every vertex yields a constraint problem, which we solve with an SMT solver. The output partitions the creases into *folds*, sets of creases that must be made together, and orders the folds partially. We work two small patterns by hand and a 278-crease dragon by machine; on 48 patterns from a public collection, 42 are resolved, and the rest mark where the method stops.

1 Introduction

Fold a square flat, unfold it, and look at the creases. Each crease is a segment bent one of two ways, mountain or valley, and the labelled drawing is a *crease pattern*. Designers publish crease patterns instead of step-by-step instructions because one picture records the whole design. It does not record the order of the folds. We ask how much of that order can be read back from the pattern alone:

(Q1) Which creases must have been made by the same move?

(Q2) Which must have been made before which others?

The answer rests on one geometric fact (Lemma 4): when a fold passes through several layers, the creases it leaves in adjacent layers are mirror images across the crease that separates them. We apply this fact at one vertex at a time. At a vertex v, a *local history* is a sequence of folds that produces exactly the creases at v, where each fold is made through one line in the paper as it is folded at that moment (Section 3). Theorem 11 characterises local histories by a parity condition and a *wedge condition*: two creases made by the same fold are related by the product of reflections in the earlier creases between them. This turns the history at a vertex into arithmetic

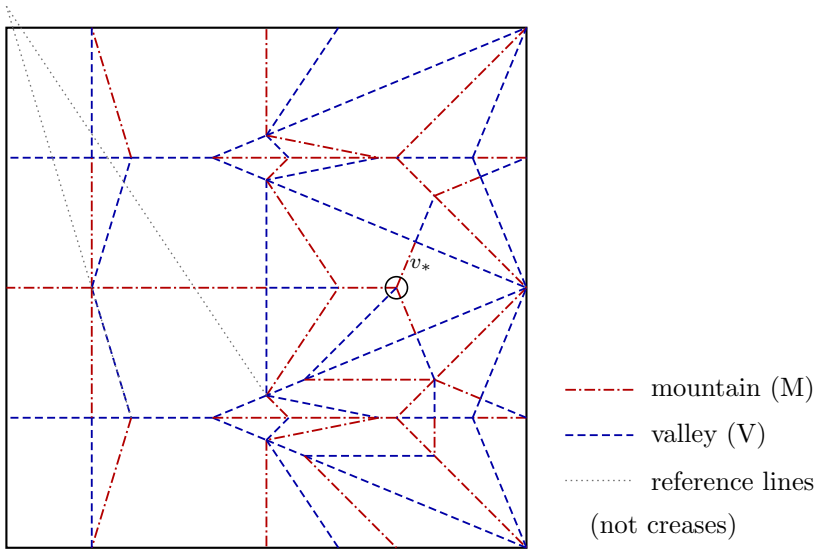


Figure 1: The running pattern, on the square $Q = [0, 8]^2$. It has 84 creases and 32 interior vertices; every crease direction is a multiple of $\pi/32$. The dotted reference lines were used to place two creases and are not part of the pattern. The circled vertex $v_* = (6, 4)$ is where a crease appears to stop for no reason (Proposition 19).

on directions. A collision test (Lemma 7) removes histories in which paper would pass through paper.

Two further ingredients complete the model. A fold that lands exactly on an existing crease erases it (Lemma 5); the erased segment leaves no trace in the pattern but is needed to explain it. We recover such *cancelled segments* by requiring that they, too, have local histories wherever they meet the pattern (Section 5). And some moves fold several lines at once; we admit them at a cost, measured by the number of distinct fold lines they use (Section 7).

The global problem asks for one order of all creases, visible and cancelled, whose restriction to every vertex is a local history (Section 6). Its solutions are the *resolutions* of the pattern. From the set of resolutions we read off the *folds*, creases that are simultaneous in every resolution, which answer (Q1) from below, and the *forced order*, which answers (Q2).

Everything here is a necessary condition for foldability by the moves we allow. We do not compute the order of layers in the folded stack, so a resolution is evidence for a folding sequence, not a certificate.

We work two small patterns by hand. The running pattern (Figure 1) has 84 creases. One of its vertices has no local history, and the theory repairs it with a single cancelled segment that it finds without being told where to look. The body pattern (Figure 3) has 15 creases. It shows the collision test at work: a partition of the creases that satisfies every wedge condition is ruled out because paper would pass through paper. Section 8 describes the computation, Section 9 applies it to a dragon with 278 creases, and Section 10 reports on 48 patterns from a public collection. Section 11 lists what remains open.

2 Folding, reflection and cancellation

Definition 1. A *crease pattern* on a polygon Q is a finite set of closed segments in Q (the *creases*) that meet only at endpoints. The endpoints are *vertices*; a vertex is *interior* if it lies in the interior

of Q. The creases divide Q into convex polygons, the *faces*.

We work with the geometric shadow of folding: where each point of the paper goes, not how the layers are stacked.

Definition 2. A *flat folded state* of a crease pattern is a continuous map $f\colon Q \to \mathbb{R}^2$ that is an isometry on each face and satisfies, for faces F, F' sharing a crease e,

$$f|_{F'} = f|_F \circ \rho_e, \tag{1}$$

where ρ_e is the reflection of the plane in the line through e.

Equation (1) says that the paper turns over at every crease.

Definition 3. A *single-line move* along a line ℓ in the plane replaces f by $f' = \rho_\ell \circ f$ on a union M of faces (the *moving* paper, after cutting faces along $f^{-1}(\ell)$ where necessary) and leaves f unchanged elsewhere. Its new creases lie in $f^{-1}(\ell)$, exactly where moving and staying paper meet.

Lemma 4 (Reflection)**.** *Let F, F' be faces sharing a crease e, and suppose a move along ℓ makes new creases in both. Then inside $F \cup F'$ these creases are mirror images of each other in the line of e.*

Proof. The new crease in F' lies in $(f|_{F'})^{-1}(\ell) = (f|_F \circ \rho_e)^{-1}(\ell) = \rho_e\big((f|_F)^{-1}(\ell)\big)$. □

Lemma 5 (Cancellation)**.** *Let e be a crease between faces F, F' with $f(e) \subset \ell$. If a move along ℓ moves F' but not F, then afterwards e is no longer a crease.*

Proof. Since $f|_F$ carries the line of e onto ℓ, $f|_F \circ \rho_e = \rho_\ell \circ f|_F$. Hence $f'|_{F'} = \rho_\ell \circ f|_{F'} = \rho_\ell \circ f|_F \circ \rho_e = f|_F \circ \rho_e \circ \rho_e = f'|_F$, so (1) no longer holds across e: the paper is flat there. □

A fold that lands exactly on an earlier crease therefore erases it, as does unfolding. We think of the erased segment as carrying two creases, one from each move, that cancel.

Lemma 6 (Moving boundary)**.** *Let F, F' be faces sharing an edge e. If a move along ℓ moves F' but not F, then $f(e) \subset \ell$.*

Proof. A point $x \in e$ ends at $f(x)$ as a point of F and at $\rho_\ell(f(x))$ as a point of F'. Continuity forces $\rho_\ell(f(x)) = f(x)$. □

For the collision test we add one physical assumption: in a single-line move, each moving face turns rigidly through a half-turn about ℓ, and paper cannot pass through paper.

Lemma 7 (Collision)**.** *Let F and G be faces, neither creased by a single-line move along ℓ, and let ρ be a ray of ℓ such that $f(F)$ and $f(G)$ each contain, in their interiors, all points of ρ close enough to its endpoint. Then F and G both move or both stay.*

Proof. Suppose F moves and G stays. Near the endpoint of ρ, the images of F and G overlap on both sides of ℓ, so in the stack one of them lies above the other there. In the half-turn of F about ℓ, the points of F on one side of ℓ leave the plane upwards and those on the other side leave it downwards, whatever the sense of rotation. One of these two halves therefore sweeps through G immediately (Figure 5(b)). □

3 Local histories

3.1 Directions and folding maps at a vertex

Fix an interior vertex v. A crease at v leaves v along a ray, and we identify the ray with its direction $x \in \mathbb{R}/2\pi\mathbb{Z}$, measured counterclockwise from the positive x-axis. Two rays lie on one line exactly when their directions agree modulo π. Reflection in a line of direction c acts on directions by

$$\sigma_c(x) = 2c - x,$$

which depends only on c modulo π. Every isometry φ of the plane acts on directions as $x \mapsto \pm x + o$, and such maps respect congruence modulo π. If φ is such a map and x a direction, then

$$\varphi \circ \sigma_x \circ \varphi^{-1} = \sigma_{\varphi(x)}, \tag{2}$$

as one checks in the two cases $\varphi(x) = x + o$ and $\varphi(x) = -x + o$.

A *state* at v is a finite set $S = \{s_1 < \cdots < s_m\}$ of rays, listed counterclockwise from some starting ray. Its *sectors* are the open wedges (s_j, s_{j+1}), indices modulo m; if S is empty there is one sector, the whole circle. The *folding maps* of S are

$$\varphi_1 = \mathrm{id}, \qquad \varphi_{j+1} = \varphi_j \circ \sigma_{s_{j+1}},$$

with φ_j belonging to the sector that starts at s_j. They describe a flat folded state near v exactly when going once around returns to the identity, $\varphi_m \circ \sigma_{s_1} = \mathrm{id}$; we then call S *flat*. This is Kawasaki's condition: m is even and alternate angles sum to π [7]. For a direction x in the sector of φ_j, the *image* of x is $\varphi_j(x)$: the direction, in the folded state, of a ray from v in direction x.

3.2 Steps and histories

By Lemma 4, a single-line move through the image of v creases the paper near v along rays whose images lie on the fold line, and by Lemma 6 the paper switches between moving and staying exactly at those rays.

Definition 8. Let S be a flat state. A *single-line step* from S is a set T of rays not in S such that

(P) $|T|$ is even;

(L) the images of the rays of T under the folding maps of S lie on one line;

(C) colouring the sectors of $S \cup T$ alternately moving and staying, switching at each ray of T, no two sectors of S that contain no ray of T and contain preimages of the same ray of that line receive different colours.

The new state is $S \cup T$.

Condition (P) is the parity of the switches in Lemma 6, (L) is Lemma 4, and (C) is Lemma 7 applied to the sectors near v. A sector bounded by a ray of T has one edge of its image on the fold line and cannot straddle it, which is why (C) concerns only uncut sectors of S.

Definition 9. A *local history* of a set A of rays is a sequence of single-line steps $T_1, \ldots, T_k$ from the empty state, with $T_1 \cup \cdots \cup T_k = A$. Equivalently, it is a weak order on A whose classes, taken in order, are single-line steps. A weak order satisfying (P) and (L) at every step is a *history up to collisions*.

Section 5 extends histories by steps that erase rays.

Lemma 10 (Flat extension)**.** *If S is flat and T satisfies (P) and (L) with respect to S, then $S \cup T$ is flat.*

Proof. Let λ be the line of the images in (L). Walk once around v, starting in a sector of $S \cup T$ that is contained in the first sector of S. Crossing a ray $s \in S$ in a sector of map φ multiplies the map on the right by σ_s, as before. Crossing a ray $x \in T$ inside a sector of S with map φ multiplies on the right by σ_x, and $\varphi\sigma_x = \sigma_{\varphi(x)}\varphi = \sigma_\lambda\varphi$ by (2). By induction the map after a crossing is $\sigma_\lambda^c\varphi$, where φ is the corresponding map of S and c is the number of rays of T crossed so far. Going all the way round gives $\sigma_\lambda^{|T|} \circ \mathrm{id} = \mathrm{id}$, because $|T|$ is even. □

3.3 The wedge characterisation

For rays $x \neq y$ and a state S, write $W_S(x, y) = (w_1, \ldots, w_r)$ for the rays of S strictly inside the counterclockwise wedge from x to y, listed counterclockwise.

Theorem 11 (Wedge characterisation)**.** *A weak order on a set A of rays at v, with classes $T_1 < \cdots < T_k$ and $S_j = T_1 \cup \cdots \cup T_{j-1}$, is a history up to collisions if and only if every $|T_j|$ is even and, for all $x, y \in T_j$,*

$$y \equiv \sigma_{w_r} \circ \cdots \circ \sigma_{w_1}(x) \pmod{\pi}, \qquad (w_1, \ldots, w_r) = W_{S_j}(x, y). \tag{3}$$

The product has the closed form

$$\sigma_{w_r} \circ \cdots \circ \sigma_{w_1}(x) = \begin{cases} 2s - x & r \text{ odd}, \\ x + 2s & r \text{ even}, \end{cases} \qquad s = w_r - w_{r-1} + \cdots + (-1)^{r-1} w_1. \tag{4}$$

Proof. Assume S_j is flat, with folding maps φ. If x lies in the sector with map φ_x and y in the sector with map φ_y, walking counterclockwise from x to y crosses exactly $w_1, \ldots, w_r$, so $\varphi_y = \varphi_x \circ \sigma_{w_1} \circ \cdots \circ \sigma_{w_r}$. The images of x and y lie on one line if and only if $\varphi_x(x) \equiv \varphi_x\sigma_{w_1} \cdots \sigma_{w_r}(y)$ (mod π). Since φ_x and the σ's respect congruence modulo π and the σ's are involutions, this holds if and only if (3) holds. Condition (L) for T_j is (3) for all pairs in T_j, because lying on one common line is transitive.

Now argue by induction on j. The empty state is flat. If S_j is flat and T_j satisfies (P) and (3), it satisfies (P) and (L), so S_{j+1} is flat by Lemma 10. Conversely, a history up to collisions has flat states throughout by the same lemma, and the computation above gives (3). Finally, (4) follows by induction on r from $\sigma_b\sigma_a(x) = x + 2(b - a)$. □

The condition involves only the earlier creases in one wedge (Figure 2). Because S_j is flat, the clockwise wedge gives the same condition.

Corollary 12. *In a history up to collisions:*

1. *two rays made together with no earlier ray between them are collinear, $y = x + \pi$;*
2. *two rays made together with exactly one earlier ray w between them satisfy $y \equiv \sigma_w(x)$: the second is the first reflected in w;*
3. *the first step is a single straight line through v;*
4. *a vertex with no two collinear rays has no local history.*

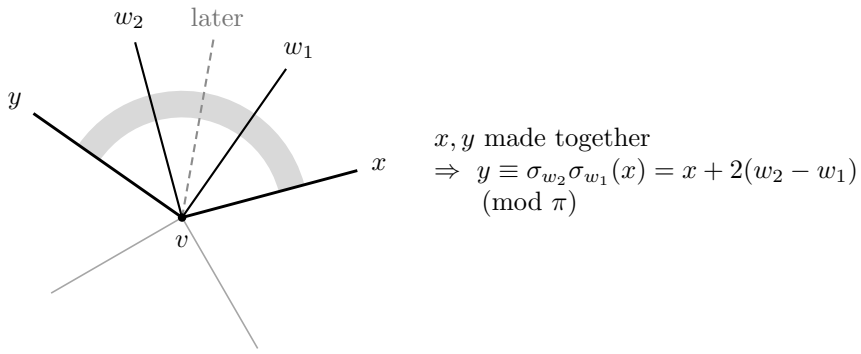


Figure 2: The wedge condition. Only the earlier creases strictly inside the counterclockwise wedge from x to y (shaded) enter; later creases and creases outside the wedge do not.

Proof. (1) and (2) are (4) with $r = 0$ and $r = 1$. (3) is (1) with $S_1 = \emptyset$, together with the fact that a line meets v in at most two rays. (4) follows from (3). □

Items (1) and (2) are the two ways a chain of creases can continue through a vertex: straight on, or bent across one earlier crease. The theorem says that these are special cases of a single rule, in which a fold may bend across any number of earlier creases at once.

Example 13 (A crossing). In the running pattern, measure directions in units of $\pi/32$, so that directions are integers modulo 64 and lines are integers modulo 32. At $(6 - 2\sqrt{2}, 6) \approx (3.172, 6)$ the rays are $0, 4, 32, 60$. By Corollary 12(3) the first step is $\{0, 32\}$, the horizontal A^+. For $\{4, 60\}$ the counterclockwise wedge from 60 to 4 contains the earlier ray 0, and $60 + 4 \equiv 2 \cdot 0 \pmod{32}$, so the second step is valid. Starting instead with $\{4, 60\}$ fails, because these rays are not collinear. The vertex has exactly one local history: A^+, then the chain P_1^+ bending across it.

Example 14 (Three folds at one point). At $(4, 8-4\sqrt{2}) \approx (4, 2.343)$ the rays are $4, 10, 16, 36, 56, 62$. The first step must be a line, and only $\{4, 36\}$ is one; it belongs to the fold P_1^-. Next, $\{16, 56\}$ has the single earlier ray 36 in its wedge, and $16 + 56 \equiv 2 \cdot 36 \pmod{32}$. The pair $\{10, 62\}$ has 36 in its wedge, and also 16 and 56 if $\{16, 56\}$ came first. In the first case $10 + 62 \equiv 2 \cdot 36$; in the second $s = 56 - 36 + 16 = 36$ and again $10 + 62 \equiv 2s$. So $C = \{16, 56\}$ and $C' = \{10, 62\}$ may come in either order after P_1^-. They cannot be made together, because their images under the fold along P_1^- are the distinct lines 16 and 10. This vertex has two local histories.

4 Collisions at a vertex

Condition (C) is not implied by (P) and (L). The body pattern (Figure 3) shows this. Its directions are multiples of $\pi/4$; in this section we measure them in units of $\pi/4$, so directions are integers modulo 8 and lines integers modulo 4.

Proposition 15. *The centre $c = (4, 4)$ of the body pattern, with rays $1, 2, 3, 5, 6, 7$, has exactly two local histories: the vertical $X = \{2, 6\}$, then $V^+ = \{1, 3\}$ and $V^- = \{5, 7\}$ in either order. Two further weak orders are histories up to collisions: the diagonals $D_1 = \{1, 5\}$ and $D_2 = \{3, 7\}$ in either order, then X. Condition (C) rejects both.*

Proof. By Corollary 12(3) the first step is one of the three lines X, D_1, D_2.

After X, a diagonal cannot be made: the wedge from 1 to 5 contains the earlier ray 2, and $1 + 5 \not\equiv 2 \cdot 2 \pmod 4$, and similarly for $\{3, 7\}$, whose wedge contains 6. The pair $\{1, 3\}$ has the

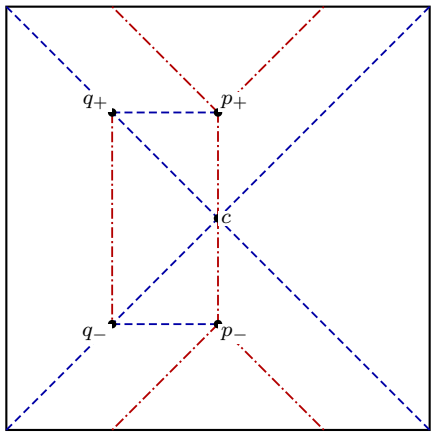
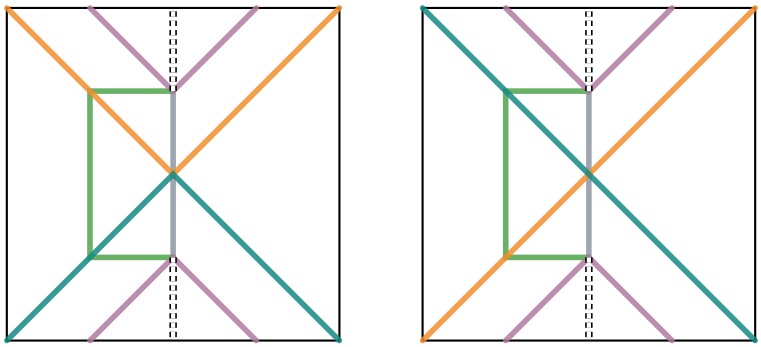


Figure 3: The body pattern on $Q = [0, 8]^2$, drawn with the conventions of Figure 1. Its interior vertices are the centre c, the points $p_\pm$ on the central vertical, and the corners $q_\pm$ of the rectangle.



A B

Figure 4: Two partitions of the creases of the body pattern into folds, with the cancelled segments above p_+ and below p_- (double dotted). Grey: the vertical X; green: the rectangle R; mauve: the small V's U and L; orange and teal: the big V's V^+ and V^- (left) or the diagonals D_1 and D_2 (right). Both satisfy the wedge conditions at c; only the left one survives the collision condition (Proposition 15).

earlier ray 2 in its wedge, and $1 + 3 \equiv 2 \cdot 2$, so it is valid; similarly $\{5, 7\}$ with 6. The two V's cannot be made together, because their images are the distinct lines 3 and 1. Condition (C) holds for these steps: the moving paper lies on one side of the fold line, as when an ordinary flap is folded.

After D_1, the line D_2 is valid, since its wedge from 3 to 7 contains the single earlier ray 5 and $3 + 7 \equiv 2 \cdot 5 \pmod 4$. The same holds after D_2. Then X has the earlier rays $3, 5$ in its wedge, with $s = 5 - 3$, and $6 - 2 \equiv 2s \pmod 4$. So both orders pass (P) and (L).

For (C), fold X after both diagonals. The four sectors E, N, W, S between the diagonals are stacked on one quarter-turn wedge, and both halves of the vertical land on its bisector, which is therefore the fold line (Figure 5(a)). The sectors E and W are not cut by the new rays, yet both contain preimages of the same ray of the fold line. The moving paper is one whole side of the vertical, so exactly one of E, W moves. Condition (C) fails, and Lemma 7 shows why (Figure 5(b)). □

The partition into X, D_1, D_2 is therefore excluded at c itself (Figure 4, right), and no global reasoning is needed to exclude it. Folding all three lines at once would produce it, but that is a

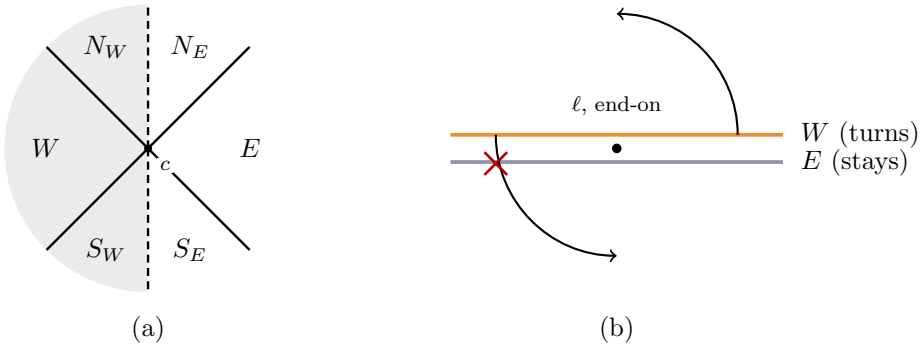


Figure 5: (a) The state at c in partition B just before the vertical (dashed) is folded: the diagonals divide the paper into sectors E, N, W, S, and the vertical cuts N and S in half. The moving paper is one whole side of the vertical, for instance the shaded one. (b) Cross-section through the stack, looking along the fold line ℓ. The sector W straddles ℓ and turns over about it while E, which straddles the same ray, stays. The two halves of W leave the plane in opposite directions, so one of them passes through E (cross).

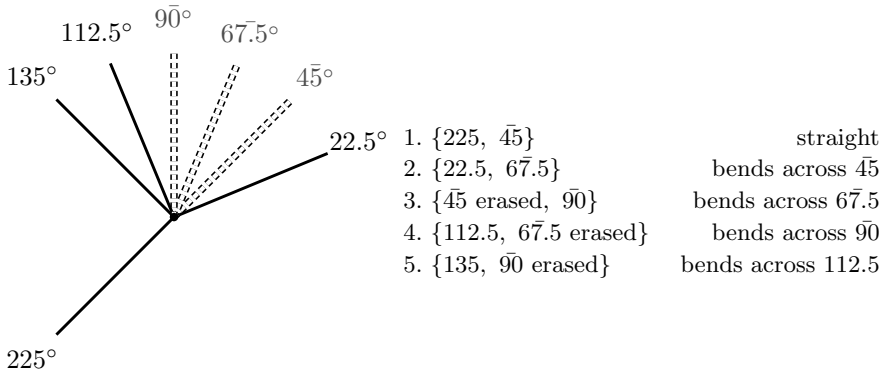


Figure 6: A staircase of cancellations at a vertex of Komatsu's Horned Owl. Solid: the four visible creases. Double dotted: three hidden rays, each created by one step and erased by a later one. Each step bends across the ray made just before it.

move through several lines, which Section 7 treats separately.

5 Cancellation

5.1 Erasing steps

By Lemma 5, a move can erase an existing crease, and the erased crease does not appear in the pattern. At a vertex this gives a second kind of step. A *hidden ray* is a ray created by one step and erased by a later one; it carries two creases, the *first* and the *cancelling* crease. A step may now also contain rays of the current state that are hidden: such rays are switch rays for (P), (L) and (C), and they leave the state. Visible rays are never erased. A *local history with cancellations* of a set A of visible rays is a sequence of such steps from the empty state that ends in the state A. Theorem 11 applies to each step unchanged, with S_j the current state. The following vertex needs three hidden rays.

Example 16 (A staircase). Take the vertex of Komatsu's Horned Owl with visible rays at 22.5°, 112.5°, 135° and 225° (Figure 6). No two are collinear, so by Corollary 12(4) there is no history

without cancellations. The history in the figure uses three hidden rays. Step 1 is a line. In each later step the counterclockwise wedge between the two switch rays contains exactly one ray of the current state, the one made just before, and the step is its reflection: $22.5 + 67.5 = 2 \cdot 45$, $45 + 90 = 2 \cdot 67.5$, $67.5 + 112.5 = 2 \cdot 90$, $90 + 135 = 2 \cdot 112.5$. Each step after the second erases the hidden ray made two steps earlier, and the state ends with the four visible rays. The exact search of Section 8.3 finds no history with fewer than three hidden rays in directions distinct from each other and from the visible rays, whatever those directions are.

5.2 The history pattern

A hidden ray at v is the end of a cancelled segment running across the sheet, and that segment must be explained wherever it meets the pattern.

Definition 17. A *history pattern* for a crease pattern $\mathcal{P}$ consists of $\mathcal{P}$ together with finitely many *cancelled segments*, each carrying a first and a cancelling crease, and an order (a weak order) on all creases, such that:

- every vertex of the combined arrangement has a local history with cancellations consistent with the order, and erases no visible crease;
- every point where cancelled creases meet, bend or end is a vertex of the combined arrangement.

The vertices of the combined arrangement include those of $\mathcal{P}$, the points where a cancelled segment crosses a crease, and its bends and ends. The next lemma settles the crossings.

Lemma 18 (Crossing)**.** *Let a cancelled segment with creases h_1 (first) and h_2 (cancelling) cross a crease z at a point q interior to both. Then the history at q is one of the following:*

1. *z is later than h_2, and the segment passes straight through q;*
2. *z is earlier than h_1, and the segment bends at q, continuing as its mirror image in z;*
3. *z is perpendicular to the segment, which passes straight through q, in any order.*

Proof. At q the visible rays are the two halves of z, made in one step, and the hidden rays are an incoming ray a and an outgoing ray b, each carrying h_1 and h_2. The steps are $\{a, b\}$ for h_1, then $\{a, b\}$ for h_2, erasing them, and the step z somewhere among them.

- If z comes first, the wedge from a to b contains one half of z, so $b \equiv \sigma_z(a)$ by Corollary 12(2).
- If z comes last, the paper at q is flat when the hidden steps are made, and $b = a + \pi$.
- If z comes between them, the step $\{a, b\}$ for h_1 is the first step at q, so $b = a + \pi$. When z is made, the wedge from one half of z to the other contains exactly one of a, b, and (3) gives $2z + \pi \equiv 2a \pmod{\pi}$, that is $2z \equiv 2a \pmod{\pi}$. Since $z \not\equiv a \pmod{\pi}$, this means $z \perp a$.

In every case with a straight segment and z perpendicular, all three orders pass. □

So a cancelled segment can be traced from either end: through a later crease it continues straight, at an earlier crease it reflects, and a perpendicular crease leaves it unchanged. A cancelled segment ends at the boundary, or at a vertex of $\mathcal{P}$ whose local history contains the arriving hidden ray.

Proposition 19. *The vertex $v_* = (6, 4)$ of the running pattern, with rays $12, 32, 40, 52$ in units of $\pi/32$, has no local history without cancellations. With one hidden ray at v_*, there is exactly one: the hidden ray $\bar{0}$ (direction 0), and the history*

$$\{32, \bar{0}\} \;\to\; \{12, 52\} \;\to\; \{40, \bar{0} \text{ erased}\}.$$

The cancelled segment runs from v_* to the boundary point $(8, 4)$ without meeting another crease.

Proof. No two rays at v_* are collinear, so Corollary 12(4) excludes histories without cancellations. With one hidden ray h, every step that contains h contains a visible ray as well, since steps have even size. So h is a preimage of the image line of a visible ray, and the enumeration of Section 8.2, which tries every such preimage, is complete here. It leaves only the history shown. Its steps can be checked by hand.

- The first step is a line, 32 and 0.
- The wedge from 12 to 52 contains the earlier ray 32 and the later ray 40; $12+52 = 64 \equiv 2\cdot 32 \pmod{32}$.
- The wedge from 40 to $\bar{0}$ contains only 52; $40 + 64 = 104 \equiv 2 \cdot 52 \pmod{32}$.

The ray from $(6, 4)$ in direction 0 reaches $(8, 4)$ without meeting a crease. □

Read as folding history: the horizontal crease A^0 along $y = 4$ once ran to the edge, the fold I was made across it, and later the fold F landed exactly on its right-hand part and erased it. The vertex was found because it has no history; the direction of the erased segment was forced.

6 Resolutions, folds and the forced order

Definition 20. A *resolution* of a crease pattern $\mathcal{P}$ is a history pattern for $\mathcal{P}$ (Definition 17). Its *geometry* is its set of cancelled segments. Its *cost* is the pair (κ, η), compared lexicographically: κ is the total cost of the collapse steps of Section 7 over all vertices (0 if there are none), and η is the number of ends of cancelled segments that lie at vertices of $\mathcal{P}$. A resolution is *minimal* if no resolution has smaller cost.

A segment that joins two vertices of $\mathcal{P}$ counts twice, once at each end, and a segment that passes through a vertex of $\mathcal{P}$ is not counted there.

Fix a geometry. Every order that makes it a resolution determines, at each vertex, which creases are made in the same step. We read the answers to (Q1) and (Q2) off what all these orders have in common.

Definition 21. Two creases at a common vertex are *forced together* if they are made in the same step in every resolution with the given geometry. The *folds* are the connected components of the relation "forced together". A fold K is *forced before* a fold K' if every crease of K is earlier than every crease of K' in every such resolution.

Proposition 22. *In any resolution with the given geometry, the creases made by one move form a union of folds, and the forced order is a partial order on the folds.*

Proof. Creases forced together share a time in every resolution, and so do all creases of a fold, by connectedness. A move is a set of creases sharing a time, so it is a union of folds. The forced order is irreflexive and transitive, because it is the intersection of the strict orders that the resolutions induce on folds. □

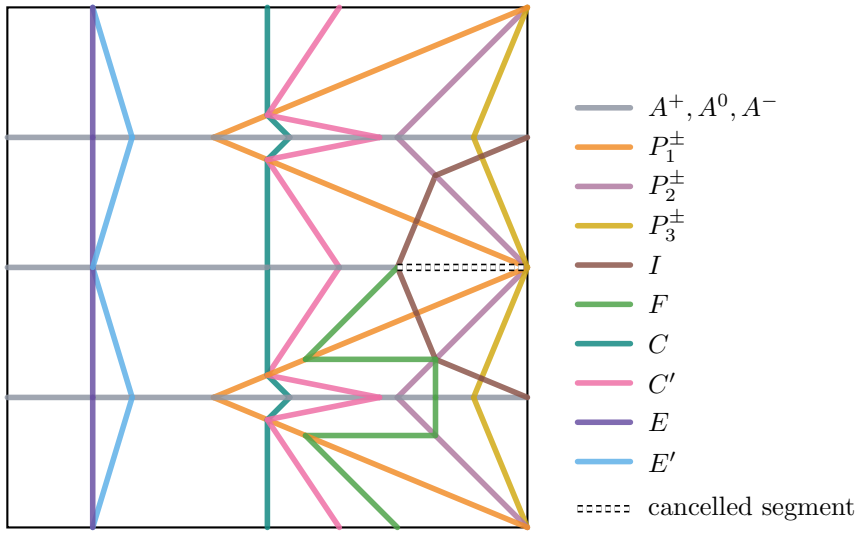


Figure 7: The 15 folds of the running pattern in its unique minimal resolution. A superscript + marks the upper member of a pair of folds that are mirror images in the line $y = 4$. The double dotted segment from $v_* = (6, 4)$ to $(8, 4)$ is the cancelled segment: its first crease belongs to A^0, its second to F.

Folds answer (Q1) from below. A fold is made by one move, but one move may make several folds, for instance the two mirror-image creases of a fold through a stack. Merging folds into moves requires global geometry, which we discuss in Section 11.

Theorem 23. *The running pattern has no resolution without cancelled segments. It has exactly one minimal resolution, whose geometry is the single cancelled segment from v_* to $(8, 4)$. In it:*

1. *there are* 15 *folds (Figure 7): the horizontals A^+, A^0, A^- along $y = 6, 4, 2$; three pairs $P_1^\pm, P_2^\pm, P_3^\pm$ of mirror-image folds at the right end of the sheet; I and F, which meet at v_*; the folds C, C' near $x = 4$; and E, E' near $x = 1.3$;*

2. *the forced order has* 30 *relations, its Hasse diagram has the* 19 *edges of Figure 8, and it has* 172,768,020 *linear extensions;*

3. *the three horizontals are minimal, and A^0 is forced before I, which is forced before F;*

4. *E is incomparable with every other fold.*

Proof. By Corollary 12(4) and Proposition 19, v_* needs a hidden ray, and with one hidden ray its history is unique. The other 31 interior vertices have local histories without cancellations, listed in Appendix A. The cancelled segment from v_* meets no crease, so it adds no vertex, and the minimal cost is one hidden ray. The folds and the forced order are computed as in Section 8.5, and the count of linear extensions by dynamic programming over subsets of folds. □

Item 4 has a simple cause. The fold E is the vertical line $x = 8\sqrt{2} - 10 \approx 1.313$, and it meets other creases only at right angles. At a vertex where two perpendicular lines cross, either order passes Theorem 11, because the wedge of the second line contains one earlier ray w with $x + (x + \pi) \equiv 2w \pmod{\pi}$ exactly when $w \perp x$. Right-angle crossings carry no ordering information, and they account for much of the large number of linear extensions.

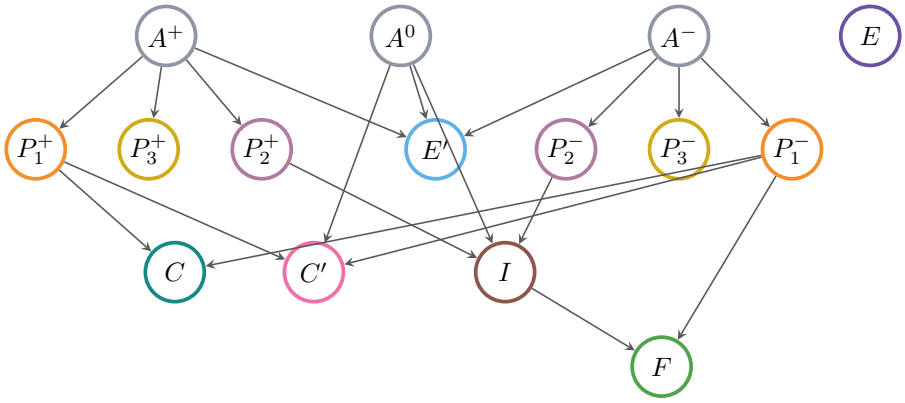


Figure 8: Hasse diagram of the forced order on the folds of the running pattern. An arrow $K \to K'$ means that K is made before K' in every resolution with this geometry.

Theorem 24. *The body pattern has exactly one minimal resolution. Its geometry is two cancelled segments, continuing the central vertical from p_+ to $(4, 8)$ and from p_- to $(4, 0)$. It has 6 folds, partitioned as on the left of Figure 4, and the forced order is generated by*

$$X \prec U,\ L,\ V^+,\ V^- \prec R,$$

with 9 relations, 8 Hasse edges and 24 linear extensions.

Proof. The vertices $p_\pm$ have no collinear rays, so each needs a hidden ray. Locally, two directions work at p_+: the vertical continuation, and a diagonal into the sheet; the same holds at p_- by symmetry. Only the vertical continuations extend to a resolution. At c the histories are those of Proposition 15. At p_+ the history is: the vertical X with the first hidden crease, then the small V U bending across it, then R bending across U and erasing the hidden ray; p_- is symmetric. The order follows. □

In folding terms: fold the sheet in half along X, make the big and small V's through both layers in any order, then make the rectangle fold R, which erases the two ends of X.

7 Moves through several lines

Some moves fold several lines at once: squash folds, petal folds, sinks and twists. In the plane, such a move replaces f by $g_i \circ f$ on regions M_i, for isometries g_i.

Lemma 25 (Boundaries of a move)**.** *Let a move replace f by $g \circ f$ on a face F and by $g' \circ f$ on an adjacent face F', across their common edge e.*

1. *If e becomes a new crease, then $g^{-1}g'$ is the reflection in a line containing $f(e)$.*
2. *If e was a crease and remains one, then $g = g'$.*

Proof. (1) Before the move $f|_{F'} = f|_F$, and afterwards (1) requires $g'f|_{F'} = gf|_F \circ \rho_e$. Hence $g^{-1}g' = f|_F\, \rho_e\, f|_F^{-1}$, the reflection in the line $f(e)$. (2) Before the move $f|_{F'} = f|_F \rho_e$; afterwards $g'f|_{F'} = gf|_F \rho_e$. Hence $g' = g$. □

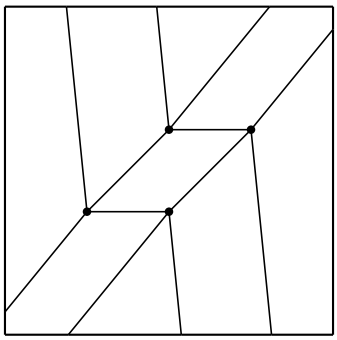


Figure 9: The 45° crossover gadget of Bern and Hayes [4], as distributed with the flat-folder collection [8] (creases unassigned). Each of its four interior vertices has sector angles 45°, 50.71°, 135°, 129.29°, in some cyclic order, and no two collinear creases.

At a vertex, then, the new creases of a move map onto a set of lines through the image of v, one for each relative isometry $g^{-1}g'$ that occurs. We call the number of these lines the *width* of the move; a single-line move has width 1. A *collapse step* at v is any set T of an even number of visible rays not yet made such that $S \cup T$ is flat. Its width w is the number of distinct lines among the images of T, and its cost is $\max(w-1, 1)$: the $w-1$ fold lines it uses beyond one, and at least 1, because a collapse step is not required to satisfy (C). Allowing collapse steps makes every flat vertex resolvable in one step, which is why they carry a cost.

Example 26 (A crossover gadget). The crossover of Figure 9 has four interior vertices, each of degree 4 with no collinear pair. By Corollary 12(4) none of them has a history without cancellations. The exact search of Section 8.3 finds none with at most four hidden rays in directions distinct from each other and from the visible rays, whatever those directions are. It was run at one vertex; the four vertices are congruent up to rotation and reflection, under which local histories are invariant. The minimal resolution the computation returns makes each vertex by one collapse of width 4, at a total cost of 12.

The example raises the main open question.

Question 27. *Does every flat vertex have a local history with cancellations, using single-line steps only?*

If the answer is yes, collapses are never necessary, only cheaper. The question is not settled by an argument about orbits of directions. A step that creates only hidden rays may use a fold line in any direction, so the directions available at a vertex are not confined to those generated by its visible rays. Example 16 shows that some vertices with no collinear rays do have such histories. An argument about the group the directions generate does, however, constrain the histories of a vertex like the crossover's. For a state S let $G(S) \subset \mathbb{R}$ be the subgroup generated by π and the differences of the directions in S.

Lemma 28. *Let a single-line step T lead from a flat state S to $S' = (S \cup T) \setminus (T \cap S)$, and let λ be its fold line. Then S' is flat and the images of all rays of T under the folding maps of S' lie on λ. If T makes at least one ray, then $G(S) \subseteq G(S')$; if it also erases at least one, $G(S) = G(S')$.*

Proof. Let $U = S \cup T$, and walk around v through the sectors of U, starting in a sector that lies in a sector of S and of S'. As in the proof of Lemma 10, crossing a ray $t \in T$ inside a sector of S with map φ multiplies on the left by $\sigma_{\varphi(t)} = \sigma_\lambda$; so the maps of U are $\sigma_\lambda^c \circ \varphi$, with φ the

corresponding map of S and c the number of rays of $T \setminus S$ crossed so far. Hence every ray of T has its image under the maps of U on λ, because σ_λ fixes λ. Crossing a ray $t \in T \cap S$ therefore also multiplies on the left by σ_λ, and the maps of U are $\sigma_\lambda^d \circ \varphi'$, with φ' the corresponding map of S' and d the number of rays of $T \cap S$ crossed. Going once around, the walk in U returns $\sigma_\lambda^{|T \setminus S|}$, since S is flat, and also $\sigma_\lambda^{|T \cap S|}$ times the return of the walk in S'; as $|T|$ is even, the walk in S' returns the identity, so S' is flat. The images of the rays of T under φ' are those under the maps of U up to a power of σ_λ, so they lie on λ.

If $x \in T$ is made by the step, then $x \in S'$, and for every $y \in T$ the computation in the proof of Theorem 11, done in S', gives $y \equiv \sigma_{w_r} \cdots \sigma_{w_1}(x) \pmod{\pi}$ with $w_1, \ldots, w_r$ the rays of S' strictly between x and y. By (4), $y - x$ is then a sum of differences of rays of S', an integer multiple of π, and twice such sums, so $y - x \in G(S')$. Every ray of S is in S' or in T, so $G(S) \subseteq G(S')$. If the step also erases a ray, the same argument with S in place of S' gives $G(S') \subseteq G(S)$. □

Proposition 29. *Let A be the rays of a flat vertex of degree 4 such that $G(A)$ is not generated by π and one other element. Then every local history with cancellations of A passes through a state with at least six rays.*

Proof. A flat state of four rays that contains a line $\{x, x + \pi\}$ has its other two rays at y and $2x - y$, one in each open half-plane, by Kawasaki's condition; so its group is generated by π and $y - x$, hence no state with group $G(A)$ contains a line. A step between two states of four rays makes as many rays as it erases, so at least one of each, and preserves the group by Lemma 28. Let S be the last state of the history with other than four rays. Every later state has four rays and group $G(A)$, so it contains no line. S is not empty, since every step from the empty state makes a line (Corollary 12(3)) and so leads to a state of two rays. Suppose $S = \{x, x + \pi\}$. Its folding maps are the identity and σ_x. For a fold line $\lambda \not\equiv x$, each of the two open half-planes contains exactly one preimage ray of λ and neither x nor $x + \pi$ maps onto λ, so a step with this line makes two rays and keeps x and $x + \pi$. For $\lambda \equiv x$, no ray in an open half-plane maps onto λ, so the step can only erase x and $x + \pi$. Either way the next state is not a state of four rays without a line. A flat state has an even number of rays, so S has at least six. □

The crossover of Figure 9 satisfies the hypothesis. Its directions are 0, $\pi/4$, θ and $\pi/4 + \theta$ modulo π, with $\tan\theta = 11/9$ by the coordinates of the file, so $G(A)$ contains $\pi/4$ and θ. If it were generated by π and some g, then $\pi/4 \in \mathbb{Z}g + \mathbb{Z}\pi$ would force $g \in \mathbb{Q}\pi$, and then $\theta \in \mathbb{Q}\pi$. But $\cos 2\theta = -20/101$ is rational and not in $\{0, \pm\frac{1}{2}, \pm 1\}$, so by Niven's theorem $2\theta \notin \mathbb{Q}\pi$. So every local history of a crossover vertex, if one exists, has a state with at least six rays, that is with at least two hidden rays present at once.

8 Computation

8.1 Directions

Crease directions are read from coordinates. We group the lines of all creases (directions modulo π) within a tolerance and replace each group by its mean. If all groups are multiples of $2\pi/N$ from the first, for some $N \leq 64$, the directions are snapped to that grid. Otherwise they are kept as real numbers. All tests use real arithmetic with a fixed equality tolerance. The tolerance starts at 10^{-6} radians and is raised, through 10^{-4}, $5 \cdot 10^{-4}$ and $2 \cdot 10^{-3}$, until every interior vertex is flat. The running pattern needs $2 \cdot 10^{-3}$: its coordinates are given to three decimals, and its directions deviate from multiples of $\pi/32$ by up to $0.061°$.

8.2 Local histories

For each interior vertex we enumerate local histories by depth-first search over steps. At a state S, the candidate fold lines are the images of the remaining visible rays, the images of the hidden rays awaiting erasure, and, for steps that create only hidden rays, the images of the bisectors of pairs of visible rays. For each candidate line, the preimage rays in the sectors of S are computed. A preimage ray that carries a remaining visible crease may be used as that crease, one that does not may become a new hidden ray, and a hidden ray already on the line may be erased. Every even subset that passes (C) is a step. Hidden rays therefore arise exactly where a fold line has a preimage that carries no visible crease, and their directions are never guessed.

The options of a vertex are grouped into *levels* (c, k), with $c \leq 1$ collapse steps and $k \leq 4$ hidden rays, ordered lexicographically; the *first level* of a vertex is the first one at which it has an option. A vertex with more than 2000 histories is not enumerated. Instead, Theorem 11 is encoded directly as constraints on the times of its creases: an even number at each time, and (3) for every pair made at the same time, with the alternating sum (4) built from comparisons between times. When the directions lie in $\mathbb{Z}/N$ with N a power of two, this arithmetic is done in fixed-width bit-vectors. Condition (C) is checked lazily: when a candidate solution violates it at such a vertex, a clause excluding that step from that earlier set is added and the solver is called again.

8.3 Exact local search

The candidate fold lines of Section 8.2 restrict only the steps that create hidden rays and no visible ones. Every other step contains a visible ray, or a hidden ray awaiting erasure, whose image fixes the fold line. For a single vertex the restriction can be removed.

Proposition 30. *Fix the visible rays at v, a number k, a cyclic order of the visible rays together with k hidden rays, and a sequence of steps, each given by the rays it makes and the hidden rays it erases. Measure each hidden direction x_i in the interval between its neighbours in the cyclic order. Then the set of $x = (x_1, \ldots, x_k)$ for which the sequence is a local history with cancellations is a finite union of sets, each defined by finitely many linear equations and strict or weak linear inequalities in x.*

Proof. The cyclic order and the sequence fix, at every step, the current state and, for every pair of rays, the rays of the state strictly inside the wedge between them. By Theorem 11 and (4), condition (L) for a step is a set of congruences $\ell(x) \equiv 0 \pmod{\pi}$, where each ℓ is affine in x with integer coefficients. Since x ranges over a bounded set, so does ℓ, and each congruence is a finite union of hyperplanes $\ell(x) = j\pi$. Flatness of each state is Kawasaki's condition, one more congruence of the same form. The folding maps of a state are $y \mapsto \pm y + o$ with o affine in x with integer coefficients, so the preimage of a ray of the fold line under the map of a piece (s, s') is an affine function $u(x)$. The piece contains it exactly when $0 < u(x) - s + 2\pi q < s' - s$, where q is the integer with $0 \leq u(x) - s + 2\pi q < 2\pi$; on each of the finitely many regions where q is constant, these are linear inequalities. Condition (C) is a Boolean combination of such conditions, and so is the conjunction of all the conditions. □

Corollary 31. *For each k it is decidable, by finitely many linear feasibility problems, whether a vertex with given visible directions has a local history with cancellations that uses k hidden rays in pairwise distinct directions different from the visible ones.*

Proof. There are finitely many cyclic orders, and finitely many step sequences, since a sequence has at most $n + 2k$ events for n visible rays. Apply Proposition 30 to each. □

We implement Corollary 31 as a depth-first search over steps for each placement of the hidden rays among the visible ones. An incremental Z3 problem over the real hidden directions and integer multiples of π discards every prefix whose conditions are infeasible. The visible directions are floating-point numbers, so the congruences are required to hold to within 10^{-9}, hidden rays are kept at least 10^{-6} from every other ray, and a preimage within 10^{-6} of the boundary of a piece does not count as inside it. Every history found is re-checked in floating point by the functions of Section 8.2.

8.4 The global problem

Each crease, visible or hidden, gets an integer time. Each vertex chooses one of its options, given by a set of hidden directions together with its histories. A chosen option requires one of its histories: equal times within a step and increasing times between steps. Each hidden ray at v requires a traced cancelled segment, following Lemma 18:

- at every oblique crossing, either the crossed crease is later than both hidden creases and the segment continues straight, or it is earlier than both and the segment continues reflected;
- the segment may end at the boundary, or at a vertex whose chosen option has the opposite hidden ray, in which case the two vertices' hidden creases are identified.

The constraints are solved with Z3 [5].

The search is started with the options of every vertex at its first level. If the constraints are unsatisfiable, the solver returns a set of vertices whose constraints conflict, and each of them is given the options of its next level as well. This repeats until a resolution exists, after which the cost is minimised over the options given so far. If (c_v, k_v) is the first level of v, every resolution in the search space (Section 8.6) has cost at least $(\sum_v c_v, \sum_v k_v)$, since a vertex at level (c, k) has c collapse steps, each of cost at least 1, and k hidden rays. When the cost found equals this *lower bound*, it is minimal over the whole search space.

8.5 Folds and the forced order

The geometry is fixed by fixing the option chosen at every vertex and, for every cancelled segment, its path: the creases where it bends, the creases and vertices it passes, and where it ends. With the geometry fixed, two creases at a vertex that share a time in a solution are forced together exactly when the solver finds no solution with different times. The folds are the components of this relation. For the forced order, every candidate pair K, K' with K earlier in a solution is tested for a solution with K' not later. Each counter-solution found discards every candidate pair it violates, so far fewer tests are needed than there are pairs.

8.6 The search space

A resolution found by the computation satisfies Definition 20. Minimality, uniqueness and the folds, however, are relative to the following restrictions, and are stated relative to them throughout.

1. Cancelled segments start at vertices of $\mathcal{P}$ and bend at most twice.
2. Where a cancelled segment passes through a vertex w of $\mathcal{P}$, every crease at w that is not perpendicular to it must be later than both of its creases. This is sufficient for a history at w, but not necessary.

3. Steps that create only hidden rays use the images of the bisectors of pairs of visible rays as fold lines. A vertex has at most four hidden rays, in pairwise distinct directions, and at most one collapse step.

4. The cost is minimised over the options given during the search. It is minimal over the search space when it equals the lower bound of Section 8.4.

5. Condition (C) is a necessary condition for physical foldability. The order of layers in the stack is not computed.

9 A larger example

Hideo Komatsu's Gentle Dragon, from the flat-folder collection [8], has 278 creases and 103 interior vertices, all directions multiples of $\pi/8$.

- *Vertices.* Eighteen vertices have no local history; fourteen of them have no collinear rays. Every other vertex has between 1 and 580 histories.

- *Resolutions.* Minimal resolutions use 24 hidden rays. Within the search space there are exactly four. They differ only at four vertices of the head, by two independent choices that are mirror images of each other. Figure 10 shows the one whose cancelled segments are shortest.

- *Cancelled segments.* Its 24 hidden rays form 15 cancelled segments.

  - Nine join two vertices that each need one. One is straight: it fills a gap in a horizontal crease near the bottom edge.
  - The other eight bend once across an earlier crease. Four of them, at the head, form two small closed loops that are mirror images of each other.
  - The remaining six run to the boundary, four straight down and two after two bends.

- *Folds and order.* There are 80 folds. Two of them consist only of cancelled creases: they are precreases that were completely erased. The forced order has 814 relations, its Hasse diagram has 115 edges, and its longest chain has 8 folds. The order is mirror-symmetric, and six folds are incomparable with all others.

The computation, including the forced order, takes about two minutes on one core.

10 A corpus

We ran the computation on 48 patterns from the collection distributed with the flat-folder program [8], with a budget of a few minutes per pattern on one core. The patterns were chosen to be small or medium-sized and varied: traditional bases and models, designs by Komatsu, Ku, Tanaka, Tachi, Furuta and others, tessellation-like patterns, and gadgets from hardness proofs [4, 2]. Table 1 groups them by what their resolution needed.

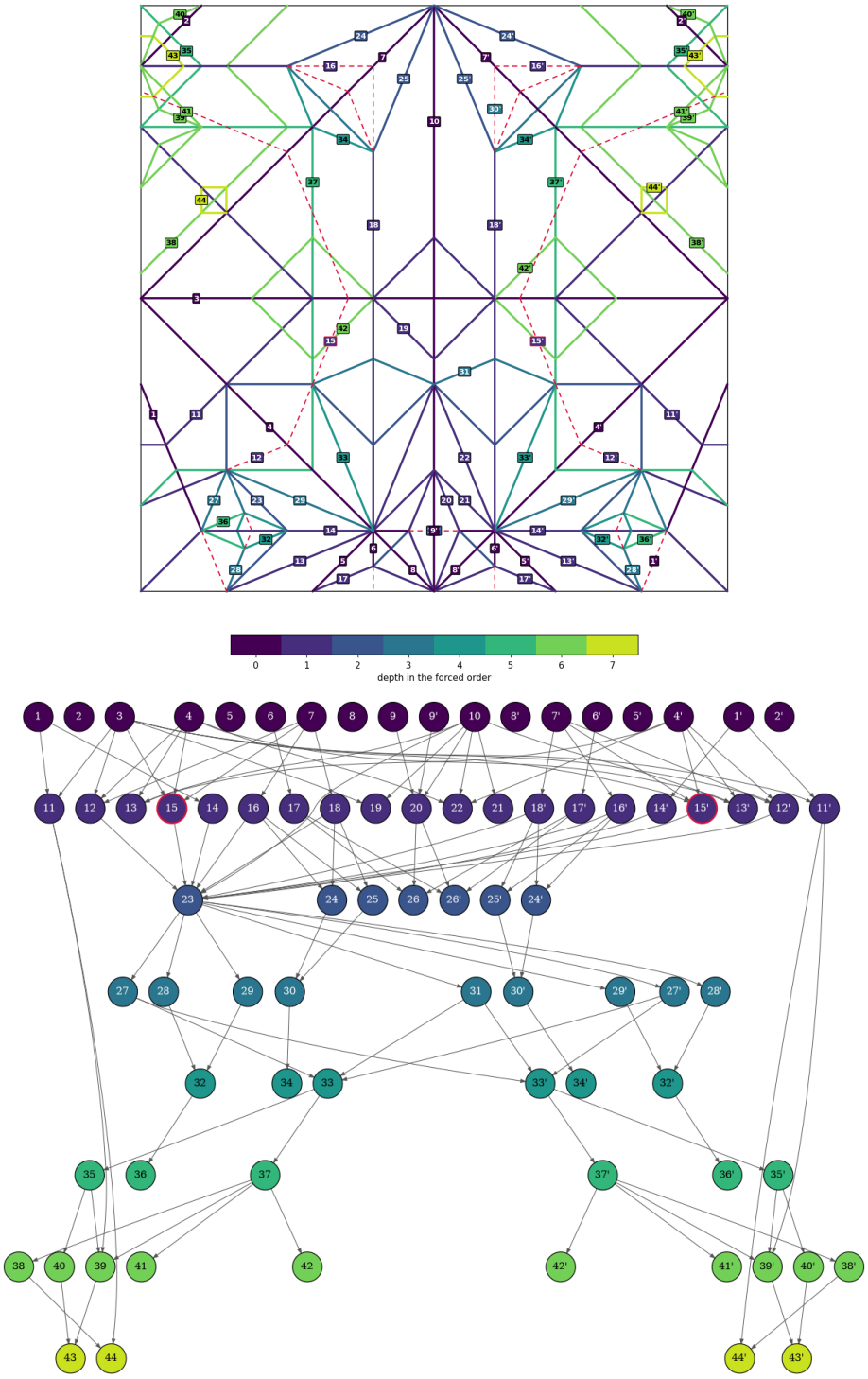


Figure 10: Gentle Dragon. Top: the 80 folds, coloured by depth in the forced order and labelled as in the Hasse diagram below; dashed crimson segments are cancelled; a prime marks the mirror image of a fold. Bottom: the Hasse diagram of the forced order. Each node is placed horizontally by its position on the sheet and vertically by its depth; crimson outlines mark folds made only of cancelled creases.

Table 1: The corpus. "Unit" is the angular unit $2\pi/N$ when one fits, and — when directions are kept as real numbers. "Hidden" is the number of hidden rays and "coll." the number of vertices made by a collapse, in the minimal resolution found. Numbers are those of the collection.

#	Pattern	Interior	Unit	Hidden	Coll.
no cancellations (8)					
001	Traditional, Sailboat	9	45°	0	0
002	Traditional, Kabuto	3	22.5°	0	0
007	Ku, Pinwheel Pockets	4	45°	0	0
026	Furuta, orihazuru	54	—	0	0
043	Ku, Quad Overlap	0	90°	0	0
074	Miura, Miura-ori	49	45°	0	0
301	boxhard, Assigned Crossover	16	90°	0	0
317	boxhard, Unassigned Crossover	4	90°	0	0
cancellations at vertices with no single-line history (20)					
004	Traditional, Crane	48	11.25°	4	0
006	Ku, Thirds Pinwheel	4	45°	4	0
038	Traditional, Yakko	16	45°	6	0
040	Komatsu, Gentle Dragon	103	22.5°	24	0
045	Traditional, Pig	21	—	4	0
052	Traditional, House	4	45°	2	0
056	Tachi, Hoodman	25	45°	2	0
062	Furuta, houou	24	—	2	0
067	Traditional, damashi bune	4	45°	2	0
073	Komatsu, Squirrel	33	22.5°	22	0
082	Traditional, 4 Birdbase	25	22.5°	16	0
089	Traditional, 9 Birdbase	69	22.5°	24	0
093	Ku, 4 Stripes 3	36	—	5	0
102	Traditional, Pinwheel	4	45°	4	0
113	Komatsu, Fox	33	22.5°	8	0
114	Kei, Kirby	20	22.5°	16	0
141	Kei, kitsune	21	22.5°	17	0
244	Komatsu, Smallbird Base	3	22.5°	2	0
316	boxhard, Unassigned Clause	3	45°	1	0
318	boxhard, Unassigned Split	3	45°	2	0
cancellations also at vertices that have one (10)					
016	Tanaka, enarc-crane final	70	11.25°	13	0
035	Tachi, kamehameha	46	45°	3	0
058	Ku, Color-change Pinwheel 1	16	22.5°	16	0
070	Ku, Pygmy Jerboa	46	22.5°	9	0
091	Ku, 4 Stripes 1	14	22.5°	6	0
096	Komatsu, Horned Owl	20	22.5°	16	0
186	Kei, Sus	18	22.5°	8	0
292	Tanaka, Aperiodic Monotile 1	10	30°	5	0
302	boxhard, Assigned Clause	23	45°	16	0
303	boxhard, Assigned Split	22	45°	6	0
collapses (4)					
015	Ku, Iso-Area Half Square	10	—	0	6
054	Tanaka, Hexagon	34	7.5°	6	2
084	Ku, 4x4 Checkerboard	20	—	14	2
325	Bern–Hayes, Unassigned 45° Crossover	4	—	0	4

#	Pattern	Interior	Unit	Hidden	Coll.
unresolved within the time budget (6)					
018	Komatsu, Butterfly Base	49	22.5°		
030	Tanaka, fish base	62	11.25°		
048	Komatsu, Dutch Rabbit	27	22.5°		
055	Komatsu, Sheep	51	22.5°		
304	boxhard, Assigned Solvable				
315	Kei, Marnie	24	22.5°		

Several patterns show the parts of the method at work:

- *Real directions.* The traditional pig uses arctan $\frac{1}{2}$ and arctan $\frac{1}{3}$, and Ku's 4×4 checkerboard uses arctan $\frac{1}{3}$; neither has an angular unit. The aperiodic monotiles use multiples of 30°.
- *Several hidden rays at one vertex.* The Horned Owl and Kirby need three or more at single vertices.
- *Hidden rays at vertices that have histories.* Ku's 4 Stripes 1, Color-change Pinwheel and the assigned split and clause gadgets of [2] need cancelled segments through vertices whose own histories need none; the conflict analysis finds them.
- *Collapses.* The crossover gadget and Ku's Iso-Area Half Square need collapses, as do Tanaka's Hexagon and the checkerboard at two vertices each.

Of the six unresolved patterns, one has more than fifteen thousand vertices. In Tanaka's fish base, the conflicts returned at each round involve 4 to 19 vertices, and expanding all of them makes each round slower than the last. The others, Komatsu's Butterfly Base, Dutch Rabbit and Sheep and Morisue's Marnie, exceed the budget for reasons we have not isolated.

11 Open problems

1. *Collapses.* Question 27: does every flat vertex have a local history with cancellations using single-line steps only? The crossover gadget is the smallest test case. A negative answer would make collapses a necessary kind of move rather than a convenience.
2. *Cancelled segments without a bend limit.* A cancelled segment's path is determined by the order: by Lemma 18 it reflects at every earlier crease it meets and passes every later one. Tracing the path implied by the current solution, and refuting it if it does not close up, would remove restriction 1 of Section 8.6 without enumerating paths. Restriction 2 can be removed in the same way, by adding the passing segment's hidden rays to the history at w.
3. *From folds to moves.* Two folds made by one move are both mapped onto the move's fold line by the folding map at that moment. Since the folding map at any time is determined by the creases present, this is a global test for merging folds into moves, and the method does not yet use it.
4. *Layers and colour.* Condition (C) is the only physical constraint used. Computing the order of layers in each intermediate state would turn resolutions into certificates and would predict the mountain and valley labels, which the method now ignores.

5. *Benchmarks.* The fish base is a search problem: shrinking each conflict to a minimal one before expanding is the obvious next step. Komatsu's Butterfly Base and Morisue's Marnie are small enough that they should be resolvable, and finding out why they are not would test restrictions 1 and 2.

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A The vertices of the running pattern

For each interior vertex of the running pattern, the table gives its coordinates, the directions of its rays in units of $\pi/32$ (with $0, 0$ for the two hidden creases at v_*), and its local histories, written as sequences of folds. Where a vertex has several histories they are separated by semicolons; the global order selects among them.

vertex	directions	local histories
$(4.000, 6.343)$	4, 10, 16, 36, 56, 62	$P_1^+ \to C' \to C$; $P_1^+ \to C \to C'$
$(1.313, 6.000)$	0, 16, 32, 48	$A^+ \to E$; $E \to A^+$
$(1.920, 6.000)$	0, 19, 32, 45	$A^+ \to E'$
$(3.172, 6.000)$	0, 4, 32, 60	$A^+ \to P_1^+$
$(4.343, 6.000)$	0, 24, 32, 40	$A^+ \to C$
$(5.725, 6.000)$	0, 30, 32, 34	$A^+ \to C'$
$(6.000, 6.000)$	0, 8, 32, 56	$A^+ \to P_2^+$
$(7.172, 6.000)$	0, 12, 32, 52	$A^+ \to P_3^+$
$(7.293, 5.707)$	4, 20, 36, 52	$I \to P_3^+$; $P_3^+ \to I$
$(4.000, 5.657)$	2, 8, 28, 48, 54, 60	$P_1^+ \to C' \to C$; $P_1^+ \to C \to C'$
$(6.586, 5.414)$	4, 24, 44, 56	$P_2^+ \to I$
$(6.293, 4.707)$	12, 28, 44, 60	$I \to P_1^+$; $P_1^+ \to I$

vertex	directions	local histories
$(1.313, 4.000)$	0, 13, 16, 32, 48, 51	$A^0 \to E \to E'$; $A^0 \to E' \to E$; $E \to A^0 \to E'$
$(4.000, 4.000)$	0, 16, 32, 48	$A^0 \to C$; $C \to A^0$
$(5.107, 4.000)$	0, 22, 32, 42	$A^0 \to C'$
$(6.000, 4.000)$	0, 0, 12, 32, 40, 52	$A^0 \to I \to F$
$(6.293, 3.293)$	4, 20, 36, 52	$I \to P_1^-$; $P_1^- \to I$
$(4.586, 2.586)$	0, 4, 8, 36	$P_1^- \to F$
$(6.586, 2.586)$	8, 20, 32, 40, 48, 60	$P_2^- \to F \to I$; $P_2^- \to I \to F$
$(4.000, 2.343)$	4, 10, 16, 36, 56, 62	$P_1^- \to C \to C'$; $P_1^- \to C' \to C$
$(7.293, 2.293)$	12, 28, 44, 60	$P_3^- \to I$; $I \to P_3^-$
$(1.313, 2.000)$	0, 16, 32, 48	$E \to A^-$; $A^- \to E$
$(1.920, 2.000)$	0, 19, 32, 45	$A^- \to E'$
$(3.172, 2.000)$	0, 4, 32, 60	$A^- \to P_1^-$
$(4.343, 2.000)$	0, 24, 32, 40	$A^- \to C$
$(5.725, 2.000)$	0, 30, 32, 34	$A^- \to C'$
$(6.000, 2.000)$	0, 8, 32, 56	$A^- \to P_2^-$
$(6.586, 2.000)$	0, 16, 32, 48	$F \to A^-$; $A^- \to F$
$(7.172, 2.000)$	0, 12, 32, 52	$A^- \to P_3^-$
$(4.000, 1.657)$	2, 8, 28, 48, 54, 60	$P_1^- \to C \to C'$; $P_1^- \to C' \to C$
$(4.586, 1.414)$	0, 28, 56, 60	$P_1^- \to F$
$(6.586, 1.414)$	16, 24, 32, 56	$P_2^- \to F$